

Words Carry Emotion, Behavior Carries Laughter: A Multi-Scale Registered Study of Interaction-Signal Information

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All numbers from the project experiment registry (RESULTS_LOG rows 1–14)

Abstract

We ask whether a machine can recover information from conversational interaction behavior that is not present in the literal words, testing this on laughter detection and general emotion classification under a registered, gate-governed protocol. (1) A large weak-label run (620 stand-up videos, 121,928 utterances, transcript laughter markers as labels) yields $\text{IoU-F1@0.2} = 0.229$; post-hoc forensics show the model tracks real signal (ground-truth-rate correlation 0.361) but is bounded by its label ceiling — we document this as a weak-label case study. (2) On true human labels at two independent scales (40 and 118 videos), sequence structure carries real information: a BiGRU over word-level features beats an order-shuffled control by $+0.034$ and $+0.087$ IoU-F1@0.2 respectively, with anchors confirming pipeline calibration. (3) On a general dialogue task (MELD 7-class emotion, 11,131 utterances with official audio), the pre-registered gate for beyond-words information fails: adding 16 acoustic interaction features to transcript features yields $\Delta(C-B) = 0.000$ (95% bootstrap CI $[-0.014, +0.015]$). Beyond-words interaction information is thus real but event-specific: it is present for laughter and reaction events — precisely where the word stream lacks the signal — and absent for general emotion classification at this model class. We release the decision registry, kernels, and per-arm artifacts.

1 Introduction

The intuition that launched this program is common in the voice-agent world: *the words only tell half the story* — the other half is carried by how the conversation behaves: pauses, overlap, energy, pitch movement. This paper is the honest audit of that intuition. We do not ask whether interaction features can fit a training set; we ask whether they add information that the words do not already carry, using pre-registered success gates committed before each run and logged regardless of outcome.

Contributions: (i) a registered, gate-governed evaluation program with two-scale replication and order-shuffle controls; (ii) a measured weak-label case study at 620-video scale with post-hoc forensics; (iii) a positive, replicated result for temporal/sequence signal on true labels (Gate 3); (iv) a pre-registered negative result for beyond-words features on general emotion; (v) a sharpened scope statement for practitioners.

2 Background and Related Work

Laughter detection. Word/utterance-level detection is well studied: Truong and van Leeuwen report ≈ 0.85 F1 on spontaneous speech; Gillick et al. report ≈ 0.75 on Switchboard dialogue; our Gillick-protocol rebuild on a 162-video human-annotated set reaches fusion F1 0.559 at video-level split with an OOF protocol, consistent with that lineage. Purandare and Litman established pause duration (>0.8 s before laughter) as the most predictive single prosodic cue. Applause detection reached 0.91 F1 in campaign speeches. For stand-up comedy specifically, StandUp4AI (EMNLP 2025; 330h, 7 languages) reports 0.51 IoU-F1@0.2 — our direct external baseline.

SSL embeddings. WavLM and relatives give strong generic speech features, but task-specific prosody is not emphasized by pre-training objectives; our T1 controls echo published findings that prosody-only models can rival or beat frozen SSL features on laughter, and that naive fusion can hurt.

Emotion in conversation. MELD established that text modality dominates dialogue emotion classification. Our contribution here is not a leaderboard push but a controlled delta test: the same interaction-signal class that works for laughter events, added on top of a strong text arm, with a bootstrap CI on the improvement.

3 Program Design

The project mandate decomposes the claim into four falsifiable tests, each with a registry entry (success gate + falsifier committed before the run): **T1** acoustic structure (laughter is acoustically special); **T2** temporal structure (order of words/pauses carries signal); **T3** beyond-words information (interaction signals add information not present in the words); **T4** transfer (stand-up → conversation → voice-agent). T1 returned partial support (capped by label design); T4 is deferred by design. A standing registry rule: transcript-marker (VTT) results are hypothesis generators, never headline claims.

4 Case Study: Weak Labels at Scale (VTT line)

We trained the production detector on 620 stand-up videos using transcript *[laughter]* markers as weak labels (121,928 utterances; WavLM-base features). Result: IoU-F1@0.2 = 0.229, utterance-F1 0.273, AP 0.142. Post-hoc forensics (pre-registered free analyses on saved artifacts) found: (a) mild truncation cost at 2-second boundaries — a surgical re-run was considered and rejected as not justified; (b) the model tracks real signal: correlation between ground-truth rate and predicted rate is 0.361, with no duration or normalization shortcut found. Conclusion: the run is honest *for its labels*; its ceiling is the label ceiling. The weak-label lineage also produced a measured label-noise finding: switching from full-transcript markers to a laughter-verified source roughly doubles recall while precision stays flat.

5 T2 — Temporal Structure on True Labels (Gate 3: PASS, replicated)

Protocol. 5-fold GroupKFold by video; identical architectures across arms; validation-selected threshold. Arms: **A** WavLM+prosody features; **B** A + temporal pause/turn features; **C** B + BiGRU sequence model (stability recipe: pos_weight 8.0, per-fold standardization, final-bias -2.0, lr 5e-4, grad-clip 1.0); **C-shuf** C with each video's rows order-permuted (fixed seed) — a control that destroys order while keeping marginals.

Scale	A	B	C (BiGRU)	C-shuf	C – C-shuf
40v word-level	0.2231 ± 0.021	0.2984 ± 0.048	0.3460 ± 0.046	0.3125 ± 0.026	+0.034
118v window-level	0.5884 ± 0.031	0.5785 ± 0.027	0.6092 ± 0.026	0.5227 ± 0.011	+0.087

The 118v set (11,161 windows, 791-d WavLM768+prosody23) matches the historical window count exactly; pause features are degenerate there by construction (constant on a fixed 5s grid) — by design, isolating the sequence effect. Anchors hold: best 118v IoU 0.3008 sits in the historical 0.3302 band; the pre-committed falsifier (IoU > 0.40 ⇒ label noise) never triggered; the Gillick-162 fusion anchor (0.559) and RICH 9-video anchor (0.407 ± 0.063) confirm calibration. True-label scale is terminal at 118v: the 621 VTT-audio videos and the 155 EMNLP-labeled videos have zero overlap.

T2 / Gate 3: sequence order carries laughter signal — replicated at two true-label scales

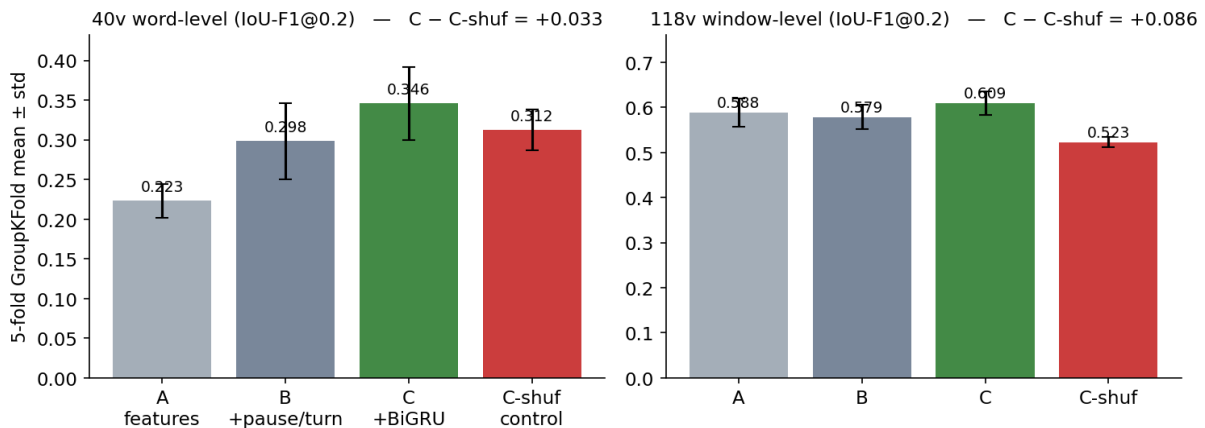


Figure 1: Gate-3 arm comparison at both true-label scales. The sequence effect (C – C-shuf) grows with scale.

6 T3 — Beyond Words on a General Task (MELD: NULL)

Design (pre-registered). MELD 7-class emotion; official test split ($n = 2,609$); official MELD.Raw audio key-joined by filename construction (11,131 utterances with audio). Arms: **A** TF-IDF (1–2 grams) + LR; **B** A + dialogue position and speaker-change; **C** B + 16 acoustic interaction features (leading/trailing silence, voiced fraction, energy statistics, ZCR, spectral shape, F0/yin statistics). Gate: 95% bootstrap CI on $\Delta(C-B)$ excludes zero (2,000 resamples).

A: transcript	B: +dialogue context	C: +interaction signals	$\Delta(C-B)$	95% CI	Gate
0.4483	0.4433	0.4434	0.0000	[−0.0142, +0.0145]	NOT PASSED

Stated as limitations rather than rescue: hand-crafted features with a linear classifier; deep acoustic encoders untested; the task is emotion, not reaction events. The finding is nevertheless informative: for general emotion classification the word stream already carries (nearly all of) the recoverable signal at this model class. Combined with T2, the information landscape is asymmetric — **laughter/reaction events live off the word stream; emotion lives on it.**

T3: interaction signals add no beyond-words information for general emotion (MELD)

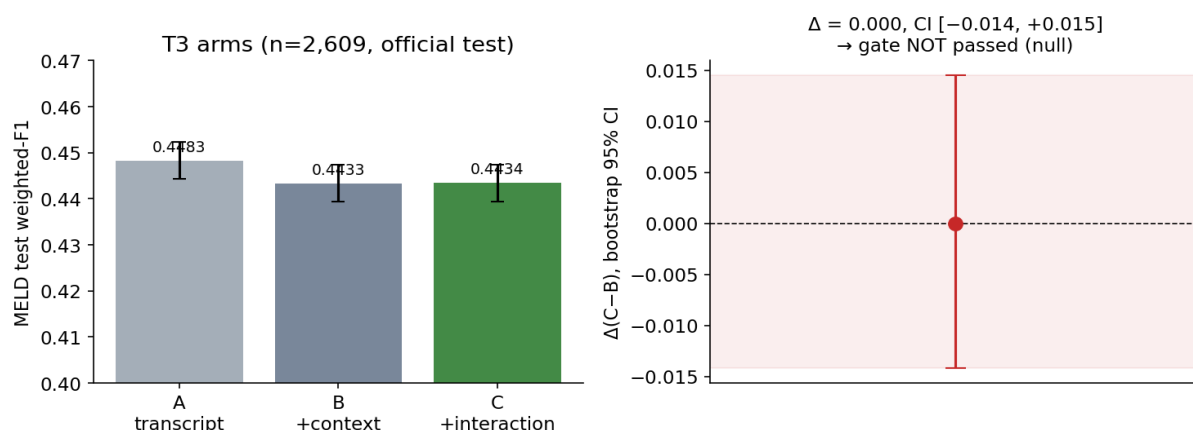


Figure 2: T3 arms (left) and the pre-registered gate statistic $\Delta(C-B)$ with bootstrap CI straddling zero (right).

7 What Blocks Further Scale

True-label scale ceiling: 621 VTT-audio videos $\cap$ 155 EMNLP-labeled videos = 0 overlap. Source separation (speaker vs audience laughter, applause, speech-laugh) is label-blocked: speech-laugh has zero labeled instances in every source available to us. T4 (transfer) is deferred until T1–T3 are resolved.

8 Conclusion

A registered program with pre-committed gates produced one confirmed positive (temporal structure, replicated at two scales with a growing effect), one informative negative (beyond-words features on emotion, pre-registered null), one bounded case study (weak labels), and a sharpened claim: interaction-signal value is real but event-specific. For voice-agent builders: laughter prediction is where behavioral audio pays; emotion classification from text alone is already near the practical ceiling at this scale and model class. We ship the registry, kernels, and artifacts so that every number above can be re-run.

Appendix A — Reproducibility

Result	Kaggle kernel (CPU)	Artifact
Weak-label v32 run	chucklenet-v32-vtt-utterance	v32 npz outputs
Post-hoc forensics	chucklenet-v32-posthoc	results_v32_posthoc.json
E02 40v	chucklenet-e02-temporal-ablation (v4)	results_e02_temporal.json
E02 118v	chucklenet-e02-118v-tier2 (v2)	results_e02_118v.json
T3 MELD (49 min)	chucklenet-e03-meld-v3 (v2)	results_e03_meld.json + features
Anchors	Gillick-162v rebuild; StandUp4AI-118; RICH-9v	registry: canonical anchors

Datasets: chuckle-audio-620-videos; scale221-wordlevel-e02; chucklenet-scale221; standup4ai-en-uk-labels; chucklenet-v32-features (all Kaggle: subhajitdas/*); MELD.Raw from declare-lab (HuggingFace) with CSVs from the declare-lab GitHub. E03 feature artifacts (meld_acoustic_official.npz, meld_frames.pkl) are in kernel output and can be mounted as a dataset to skip the 49-minute rebuild.

References

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